

Blood & Blood Vessels

What blood is made of, how it stops a leak, and why a tiny change in vessel width changes everything.

BY THE END OF THIS PAGE YOU SHOULD BE ABLE TO

- Name the parts of blood and the job of each
- Explain how oxygen is carried, and why anemia causes shortness of breath
- Walk through how bleeding is stopped, and find the positive feedback in it
- Match each vessel type to what it is built to do
- Explain why vessel width controls blood pressure and flow

1 What is in the tube

PART	SHARE	JOB
Plasma	About 55%	Water, proteins, salts, nutrients, and wastes. The delivery fluid
Red cells	About 45%	Carry oxygen, and help carry carbon dioxide back
White cells	Under 1%	Defense. See Module 11
Platelets	Under 1%	Plug damaged vessels and start clotting

Two plasma proteins do most of the work. **Albumin** holds water inside the vessels. Lose it and fluid leaks into the tissues as swelling. **Fibrinogen** is the raw material a clot is built from.

2 Oxygen is carried, not just dissolved

Very little oxygen floats loose in plasma. Almost all of it rides on hemoglobin.

Each red cell is packed with **hemoglobin**, and each hemoglobin molecule carries up to four oxygen molecules.

Red cells have no nucleus. That frees up room for more hemoglobin, but it also means they cannot repair themselves. They last about 120 days and are replaced constantly.

Now the part that matters at the bedside. A patient with anemia can have normal lungs and a normal oxygen reading and still be starved of oxygen.

The reading shows how *full* the carriers are. It does not show how *many* carriers there are. Half the trucks, all fully loaded, still delivers half the cargo.

WHY THE NURSE CARES

Iron sits at the center of hemoglobin. No iron means no hemoglobin, which means no oxygen carried. That is why iron deficiency shows up as tiredness and shortness of breath rather than as anything that sounds like a blood problem.

3 Stopping a leak, in three stages

- 1 **The vessel narrows.** This happens at once, and it slows flow through the hole.
- 2 **Platelets plug it.** They stick to the damaged wall, then release chemicals that call in more platelets, which release more chemicals. This is **positive feedback**, and it is what makes the plug form fast.
- 3 **The clot sets.** A chain of clotting factors turns fibrinogen into fibrin, a mesh of threads that traps cells and turns the soft plug into a firm clot.

Positive feedback shows up here for the same reason it shows up in labor. The body wants the job finished quickly, and there is an outside event to stop it: the sealed vessel.

Compare that with the loops holding temperature and blood sugar steady, which are built never to finish at all.

Clotting factors are made in the liver, and many need vitamin K. That one fact explains why liver disease causes bleeding, and why warfarin, which blocks vitamin K, works as a blood thinner.

4 Three vessel types, three different builds

VESSEL	WALL	BUILT FOR
Artery	Thick, stretchy, muscular	Handling pressure, and springing back to keep flow going between beats
Capillary	One cell thick	Exchange, and nothing else
Vein	Thin and floppy, with valves	Holding volume and returning blood at low pressure

Veins hold most of your blood at any moment, and they return it uphill against gravity with almost no pressure of their own.

They manage it two ways. **Leg muscles squeeze them**, and **valves stop backflow**. That is why standing still pools blood in the legs, and why moving prevents it.

5 Width rules everything

Change a vessel's width a little and watch flow change a lot.

VESSEL RADIUS, RESISTANCE AND FLOW



Radius $\times 1.00$ · resistance 100% of normal · flow 100% of normal

Resting vessel tone. Arterioles are never fully open or fully closed — they sit part way, so they can move in either direction on command.

Flow through a tube depends on the **fourth power** of its radius. Cut the radius in half and flow drops to one sixteenth.

That is why arterioles, the small vessels wrapped in muscle, are the body's main control point for blood pressure. It is also why drugs that widen or narrow them have such big effects from such small changes.

Blood pressure itself is how much the heart pumps, multiplied by how much the vessels resist. Raise either one and pressure rises.

That single sentence is behind every blood pressure drug. They either reduce what the heart puts out, or they widen the vessels it pumps into.

THINK OF IT THIS WAY

A garden hose with your thumb over the end. You barely moved your thumb, and the whole stream changed. Arterioles are the body's thumb.

6 Check yourself

Five questions. The wrong answers explain themselves, so they are worth reading.

1. A patient with severe anemia has an oxygen reading of 98%. Why can they still be short of oxygen?

- a) The reading is unreliable in anemia
- b) The reading shows how full the hemoglobin is, not how much hemoglobin there is
- c) The lungs are failing
- d) Oxygen travels dissolved in plasma

2. Which step of stopping a bleed uses positive feedback?

- a) The vessel narrowing
- b) Platelets plugging the hole
- c) The fibrin mesh forming
- d) The clot dissolving later

3. Why does liver disease cause abnormal bleeding?

- a) The liver stores platelets
- b) Clotting factors are made in the liver
- c) The liver makes hemoglobin
- d) Bile is needed to form a clot

4. Blood returns from the legs uphill at low pressure. What makes that possible?

- a) The vein walls squeeze hard
- b) Leg muscles squeeze the veins, and valves stop backflow
- c) Artery pressure pushes it all the way around
- d) Capillaries pull it back

5. An arteriole narrows so its radius is half of what it was. Roughly what happens to flow?

- a)** It halves
- b)** It drops to about a quarter
- c)** It drops to about a sixteenth
- d)** It stays the same

Answer key

1. **The reading shows how full the hemoglobin is, not how much hemoglobin there is**

Correct. Fewer carriers, all full, still deliver less oxygen.

2. **Platelets plugging the hole**

Right. Platelets call in more platelets, which call in more. The response builds on itself.

3. **Clotting factors are made in the liver**

Correct, and many of those factors also need vitamin K. That is how warfarin works.

4. **Leg muscles squeeze the veins, and valves stop backflow**

Right, and this is why standing still pools blood while walking prevents it.

5. **It drops to about a sixteenth**

Correct. Flow depends on the fourth power of radius, which is why arterioles control pressure so well.

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